# Lecture 41

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# 1 Vectors in $\mathbb{R}^n$

## 1.1 Definition

A vector in $\mathbb{R}^n$ is an n-tuple $(x_1, \ldots, x_n)$ of real numbers. Vectors correspond to points in space.

- $\mathbb{R}^1$: Real line.
- $\mathbb{R}^2$: Plane, $x = (x_1, x_2)$.
- $\mathbb{R}^3$: Space, $x = (x_1, x_2, x_3)$.
- $\mathbb{R}^n$: $x = (x_1, x_2, \ldots, x_n)$.

## 1.2 Notation

It is common in Linear Algebra to represent vectors as column vectors:

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Alternatively, using the transpose notation for space efficiency:

$$x = (x_1, x_2, \ldots, x_n)^T$$

## 1.3 Operations

Given vectors $x, y \in \mathbb{R}^n$ and a scalar $\alpha \in \mathbb{R}$:

1. **Addition**:
$$x + y = (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n)$$

2. **Scalar Multiplication**:
$$\alpha x = (\alpha x_1, \alpha x_2, \ldots, \alpha x_n)$$

### 1.4 Properties of Operations

For all $x, y, z \in \mathbb{R}^n$ and $\alpha, \beta \in \mathbb{R}$:

1. $x + y = y + x$ (Commutativity)
2. $x + (y + z) = (x + y) + z$ (Associativity)
3. $(\alpha\beta)x = \alpha(\beta x)$
4. $(\alpha + \beta)x = \alpha x + \beta x$
5. $\alpha(x + y) = \alpha x + \alpha y$

## 2 The Standard Basis and Inner Product

### 2.1 The Standard Basis

We define the special basis vectors $e_i$ where the i-th component is 1 and all others are 0:

$$e_1 = (1, 0, \ldots, 0)^T$$
$$e_2 = (0, 1, \ldots, 0)^T$$
$$e_n = (0, 0, \ldots, 1)^T$$

Any vector $x$ can be written as a linear combination of basis vectors:

$$x = x_1 e_1 + x_2 e_2 + \cdots + x_n e_n = \sum_{i=1}^{n} x_i e_i$$

### 2.2 Scalar / Dot / Inner Product

The scalar product (or dot/inner product) of two vectors $x, y \in \mathbb{R}^n$ is the number:

$$\langle x, y \rangle = x \cdot y = x^T y = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n = \sum_{i=1}^{n} x_i y_i$$

#### 2.2.1 Properties

1. $(x + y)^T z = x^T z + y^T z$
2. $x^T y = y^T x$ (Symmetry)
3. $(\alpha x)^T y = \alpha(x^T y)$
4. Vectors $x$ and $y$ are **orthogonal** if and only if $x^T y = 0$.

### 2.2.2 Norm (Length)

The length of a vector $x$ is:

$$|x| = \|x\| = \sqrt{x^T x} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

### 2.2.3 Examples of Inner Product Operations

1. Selecting a component: $e_i^T x = x_i$
2. Summing components: Let $\mathbf{1} = (1, 1, \ldots, 1)^T$. Then $\mathbf{1}^T x = x_1 + \cdots + x_n$.
3. Averaging components: $\frac{\mathbf{1}}{n}^T x = \frac{x_1 + \cdots + x_n}{n}$.
4. Zero vector: $0 = (0, \ldots, 0) \in \mathbb{R}^n$.

# 3 Hyperplanes

## 3.1 Definition

A **hyperplane** in $\mathbb{R}^n$ is the set:

$$\{x \in \mathbb{R}^n : a^T x = c\}$$

for any fixed vector $a \in \mathbb{R}^n$ ($a \neq 0$) and $c \in \mathbb{R}$. The vector $a = (a_1, \ldots, a_n)$ is the **normal vector** (orthogonal to the hyperplane).

Geometric interpretation:

- In $\mathbb{R}^2$: $a_1 x_1 + a_2 x_2 = c$ (A line).
- In $\mathbb{R}^3$: $a_1 x_1 + a_2 x_2 + a_3 x_3 = c$ (A plane).
- In $\mathbb{R}^n$: A hyperplane.

## 3.2 Example Problems

### 3.2.1 Problem 1: Parallel Hyperplane

Find a hyperplane parallel to $2x_1 - 3x_2 + x_3 - x_4 = 2$ that passes through the point $P(1, 1, 2, 2)$.

**Solution:** A parallel hyperplane has the same normal vector $a = (2, -3, 1, -1)$. We need to find the new constant $c$ by substituting the point $P$:

$$2(1) - 3(1) + (2) - (2) = c$$

$$2 - 3 + 2 - 2 = -1$$

$$c = -1$$

**Answer:** $2x_1 - 3x_2 + x_3 - x_4 = -1$.

#### 3.2.2 Problem 2: Unit Vector

Find a unit vector in the direction of $v = (5, 2, -1, 0, 1)$.

**Solution:** First, calculate the length of $v$:

$$|v| = \sqrt{5^2 + 2^2 + (-1)^2 + 0^2 + 1^2} = \sqrt{25 + 4 + 1 + 0 + 1} = \sqrt{31}$$

The unit vector is $\frac{v}{|v|}$:

$$u = \frac{1}{\sqrt{31}}(5, 2, -1, 0, 1)$$

# 4 General Vector Spaces

## 4.1 Definition

A **vector space** $V$ is a collection of elements (called vectors) along with two operations: addition of vectors and multiplication by a scalar (number), such that the following axioms hold for all $a, b, c \in V$ and $\alpha, \beta \in \mathbb{R}$ (or $\mathbb{C}$):

1. $a + b = b + a$ (Commutativity)
2. $a + (b + c) = (a + b) + c$ (Associativity)
3. There exists a **zero vector** $0 \in V$ such that $a + 0 = a$ for all $a \in V$.
4. For every $a \in V$, there exists an **additive inverse** $b \in V$ such that $a + b = 0$. Notation: $b = -a$.
5. $1 \cdot a = a$ (Identity)
6. $(\alpha\beta)a = \alpha(\beta a)$
7. $\alpha(a + b) = \alpha a + \alpha b$
8. $(\alpha + \beta)a = \alpha a + \beta a$

## 4.2 Examples

1. $\mathbb{R}, \mathbb{R}^2, \ldots, \mathbb{R}^n$.
2. $\mathbb{C}, \mathbb{C}^n$.
3. $P_n = \{p(t) \in \mathbb{R}[t] : \deg(p) \leq n\}$ (Polynomials of degree at most $n$).
4. $F = \{f : [0,1] \to \mathbb{R}\}$ (Functions mapping interval to reals).

## 4.3 Non-Examples

1. $P_n^= = \{p(t) \in \mathbb{R}[t] : \deg(p) = n\}$ (Polynomials of degree exactly $n$).
   - **Reason:** Not a vector space because it does not contain the zero polynomial (degree undefined or $-\infty$), and the sum of two degree $n$ polynomials might have degree $< n$.
2. $V = \{f : [0,1] \to \mathbb{R}^+\}$ (Non-negative functions).
   - **Reason:** Fails axiom 4. If $f(x) > 0$, then $-f(x)$ is not in the set.

## 4.4 Theorems

1. **Uniqueness of Zero**: A vector space has a unique zero vector. **Proof:** Suppose 0 and $0'$ are zeros. Then $0' = 0' + 0 = 0 + 0' = 0$.
2. **Properties**:
   - The additive inverse is unique.
   - $0 \cdot a = 0$ (Scalar zero times vector).
   - $\alpha \cdot 0 = 0$ (Scalar times vector zero).
   - $(-1) \cdot a = -a$.